

VEHICLE NUMBER PLATE SEARCH VIA BINARY SEARCH TREE (BST)

Lab Assessment / Problem Statement

PROBLEM DESCRIPTION

At an intersection, several vehicles pass by every day. We want to record the four-digit numbers appearing on their number plates and later search for specific vehicle numbers.

The vehicle numbers must be stored in a **Binary Search Tree (BST)**. Insert the vehicle numbers into an initially empty BST in the exact order in which they are given in the input.

For every search operation, determine whether the given vehicle number exists in the BST. If it exists, print the path from the root of the BST to that node:

- **L** — Move to the left child.
- **R** — Move to the right child.
- If the searched number is present at the root, the path is empty.

INPUT FORMAT

- The first line contains n distinct four-digit integers separated by a single space. These numbers must be inserted into an initially empty Binary Search Tree in the order they appear. Each number is between 1000 and 9999.
- Each subsequent line contains a search query in the format: $S \ m$, where S represents the search operation and m is the four-digit vehicle number to be searched.
- The search queries end at End Of File (EOF).

OUTPUT FORMAT

- There is no output produced after reading and inserting the numbers from the first line.
- For every search query $S \ m$:
 - If m exists, print $1 \ \text{path}$. If m is at the root, print 1 (with a trailing space).
 - If m does not exist in the BST, print 0 .
- Each result must be printed on a new line.

CONSTRAINTS

- $1 \leq n \leq 9000$
- $1000 \leq a_i, m \leq 9999$
- All vehicle numbers on the first line are distinct.
- At most 10^5 search queries.
- The BST must be constructed by inserting the numbers in the given order without balancing. Searching must be done strictly using the constructed BST.
- Each BST node must contain pointers to its left child, right child, and parent. The parent of the root should be NULL.

SAMPLE TEST CASE

Sample Input:

```
5000 3000 7000 2000 4000 6000 8000
S 4000
S 8000
S 5000
S 1234
```

Sample Output:

```
1 LR
1 RR
1
0
```

Explanation:

The BST is constructed with 5000 as root. 3000 and 7000 are its left and right children. 2000, 4000, 6000, and 8000 are leaf nodes.

- For $S \ 4000$: root $\rightarrow$ Left $\rightarrow$ Right, output is $1 \ LR$.